

Education

Carnegie Mellon University Information Networking Institute
M.S. in **Mobile and IoT Engineering**; GPA: 4.00 / 4.00

Pittsburgh, PA, USA | Aug. 2021 - Dec. 2022

Zhejiang University College of Computer Science and Technology

Hangzhou, Zhejiang, China | Sep. 2016 - Jul. 2021

B.Eng. in **Computer Science and Technology**, B.Sc. in **Statistics** (double degree); GPA: 3.63 / 4.00

Technical Skills

- **Domains:** multimodal (audio / video / motion signal) machine learning, system programming, on-device model integration & evaluation
- **Languages:** Python, Objective-C, C++, Swift, C, Shell script.
- **Frameworks & Tools:** PyTorch, CoreML, Apple Platform SDKs, Generative AI-assisted coding workflows.

Industry Experience

Apple AI/ML Software Engineer, Siri Attention & Invocation

Cupertino, CA, USA | Jan. 2023 - Present

- On-Device Multimodal Model Integration & Validation
 - Integrated a **real-time multimodal model** for Siri invocation into an **on-device trusted execution environment**. Supported cross-functional teams through feature prototyping to performance hill-climbing.
 - Connected the model inference stack to **system programming interfaces (SPIs)** for handling **streaming multimodal inputs**, including camera, audio, and pre-processed signals within a trusted execution environment.
 - Implemented **inter-process communication between the secure runtime and main OS layer** to trigger real-time device responses based on model outputs.
 - Supported feature **live-on and model optimization**, reducing false triggers by 75% at stable miss rate.
- Built an **on-device data collection framework** to enable real-world model performance analysis and validation across multiple Siri features, including voice trigger, Siri Directed Speech Detector, adaptive volume, and endpointer.
 - Developed a **file-based signal capture** within Siri's speech framework to **log raw real-time multimodal signals** (e.g., audio buffers, time-series diagnostics) during in-lab and take-home testing, enabling failure case inspection and dataset creation.
 - Integrated analytics-based signal logging and **on-device heuristics aggregation**, powering dashboards for internal carry traffic monitoring to track model quality.
- Model Evaluation Infrastructure Optimization
 - Rebuilt Siri's **voice trigger model evaluation pipeline**, migrating from x86-based emulated environment to Apple Silicon.
 - Developed a **custom evaluation binary** based on Siri's on-device inference framework, carefully reusing core model logic while adding support for accelerated sample feeding, dataset ingestion, and multi-architecture execution (application processor and always-on processor). Balanced speed and fidelity to enable fast, reproducible evaluations that reflect on-device inference behavior.
 - Designed and implemented a **distributed evaluation pipeline** using internal infrastructure to scale across virtual device fleets (iPhone, Mac, HomePod), handling workload distribution, failure recovery, and result aggregation. Cut compute usage by 10x and runtime by 2x, while closely matching real-world inference behavior.
- Initiated and prototyped a **tool based on model inversion attack to synthesize natural-sounding positive samples for the voice trigger model**. Extended the tool by migrating to internal model formats and optimizing performance. The tool was actively used by the modeling team for research, failure case analysis, and as a part of an undisclosed internal project.

Research & Internship

NVIDIA Deep Learning Infrastructure Engineer Intern

Santa Clara, CA, USA | May. 2022 - Aug. 2022

- Collaborated with machine learning researchers on a **graph convolution network (GCN)-based congestion prediction** system, improving the clustering and searching algorithm for graph generation from hardware design data and **deploying the model as async service** via a pub-sub pipeline.

Massachusetts Institute of Technology Visiting Researcher

Cambridge, MA, USA | Sep. 2019 - May. 2020

- Developed a computer vision and deep learning based food carbohydrate estimation system.
 - Trained and optimized the **semantic segmentation model for food region recognition** based on U-Net.
 - Developed the **iOS data pipeline** on raw signal capture, scene reconstruction and understanding, carbohydrate estimation, result visualization, and user interaction.
- Worked with Boston Dynamics on the healthcare-focused SPOT robot system responding to COVID-19.
 - Researched and built the **facial temperature and respiration rate monitoring system** with openCV, InsightFace, and GigE vision.
 - Configured the **CUDA environment and improved MATLAB array's memory layout compatibility** with Python, improving frame rate nearly 8 times over the baseline implementation.